

Chapter 13

The Normal Distribution

Lecture 13 | Devore 9e: 4.3

Recall the continuous toolkit from last lecture:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) dx, \quad \mathbb{E}[h(X)] = \int_{-\infty}^{\infty} h(x) f(x) dx, \quad \text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx = \mathbb{E}[X^2] - \mu^2.$$

13.1 Why this distribution and not another

Take $X_1, X_2, \dots, X_N$ independent and identically distributed — *whatever* their common distribution — and form the average

$$Z = \frac{1}{N} \sum_{i=1}^N X_i.$$

As N grows, the distribution of Z looks like the same bell-shaped curve, no matter what the X_i were. Complicated inputs, one universal output. That is the central limit theorem (Lecture 19), and it is the reason the normal distribution is everywhere: any quantity that is the accumulation of many small independent effects — measurement error, height, exam totals — comes out approximately normal.

13.2 Definition

Definition 13.1 (Normal distribution). $X \sim \text{Normal}(\mu, \sigma^2)$ if its pdf is

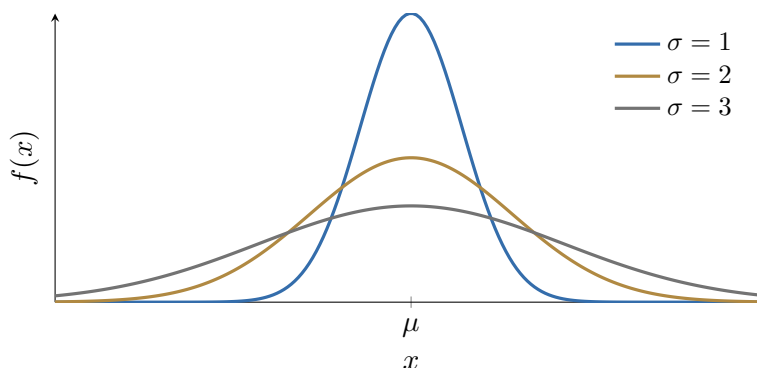
$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right), \quad x \in \mathbb{R},$$

where $\mu \in \mathbb{R}$ and $\sigma > 0$.

Three facts to check (all good exercises):

$$\int_{-\infty}^{\infty} f(x; \mu, \sigma) dx = 1, \quad \mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2.$$

The factor $\frac{1}{\sqrt{2\pi}\sigma}$ is exactly the *normalization* needed to make the total area 1; the shape is carried entirely by $\exp(-\frac{1}{2}z^2)$ with $z = \frac{x-\mu}{\sigma}$.



The curve is symmetric about μ (so the mean, median and mode all coincide there), and larger σ means more spread. Note that the peak height is $\frac{1}{\sqrt{2\pi}\sigma}$, so a very small σ gives a very tall narrow spike — again a reminder that a density is not a probability and may exceed 1.

13.3 The standard normal, and standardization

Definition 13.2 (Standard normal). $Z \sim \text{Normal}(0, 1)$, with pdf and cdf

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad \Phi(z) = \mathbb{P}\{Z \leq z\} = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$

Remark 13.3. Φ has no closed form in elementary functions — you cannot antidifferentiate $e^{-t^2/2}$ by hand. That is why the standard normal table exists, and why every normal computation is routed through it.

Theorem 13.4 (Standardization). If $X \sim \text{Normal}(\mu, \sigma^2)$ and

$$Z := \frac{X - \mu}{\sigma},$$

then $Z \sim \text{Normal}(0, 1)$.

Proof. $F_Z(z) = \mathbb{P}\left\{\frac{X-\mu}{\sigma} \leq z\right\} = \mathbb{P}\{X \leq \mu + \sigma z\} = F_X(\mu + \sigma z)$. Differentiate in z : $f_Z(z) = \sigma f_X(\mu + \sigma z) = \sigma \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-z^2/2} = \varphi(z)$. $\square$

This single fact reduces every normal question to a table lookup:

$$\mathbb{P}\{X \leq x\} = \mathbb{P}\left\{\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right\} = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

and consequently

$$\mathbb{P}\{a \leq X \leq b\} = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right).$$

Remark 13.5 (Reading the table backwards). Percentiles work the same way. If z_p solves $\Phi(z_p) = p$, then the p -th percentile of X is $x_p = \mu + \sigma z_p$. Values worth memorizing:

$$z_{0.90} = 1.28, \quad z_{0.95} = 1.645, \quad z_{0.975} = 1.96, \quad z_{0.99} = 2.33.$$

By symmetry $\Phi(-z) = 1 - \Phi(z)$, which is how negative arguments are handled.

Example 13.6. Exam scores are $\text{Normal}(72, 8^2)$. What fraction of students score between 60 and 90?

$$\mathbb{P}\{60 \leq X \leq 90\} = \Phi\left(\frac{90 - 72}{8}\right) - \Phi\left(\frac{60 - 72}{8}\right) = \Phi(2.25) - \Phi(-1.5) = 0.9878 - 0.0668 = 0.9210.$$

One more identity makes many normal expectations easy. Differentiating the density gives $\varphi'(z) = -z\varphi(z)$, so $z\varphi(z)$ has antiderivative $-\varphi(z)$ and

$$\int_a^\infty z\varphi(z) dz = \varphi(a).$$

Example 13.7 (Only the upside). A stock trades at 100 today, and its price one month from now is modeled as $X \sim \text{Normal}(100, 5^2)$. A contract pays $\max(X - 100, 0)$ dollars in one month: the gain above today's price, or nothing. What is its expected payoff?

Write $X = 100 + 5Z$, so the payoff is $5 \max(Z, 0)$. By the identity with $a = 0$,

$$\mathbb{E}[\max(Z, 0)] = \int_0^\infty z\varphi(z) dz = \varphi(0) = \frac{1}{\sqrt{2\pi}} \approx 0.399,$$

and the expected payoff is $5 \times 0.399 \approx \$2.00$, about 0.4 standard deviations. The contract is a call option, and “0.4 standard deviations” is the usual back-of-the-envelope price for one struck at today's price (ignoring interest). By symmetry $\mathbb{E}[\max(-Z, 0)]$ is the same, and $|Z| = \max(Z, 0) + \max(-Z, 0)$, so

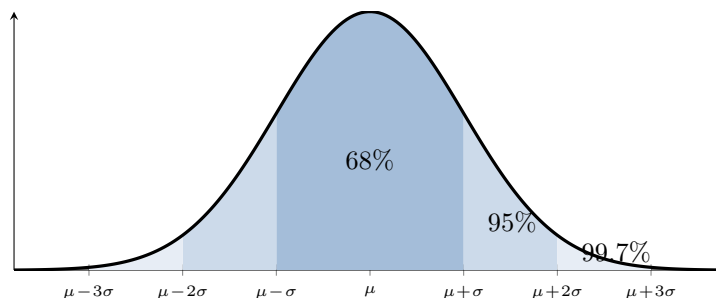
$$\mathbb{E}|Z| = \frac{2}{\sqrt{2\pi}} = \sqrt{\frac{2}{\pi}} \approx 0.80 :$$

a normal variable lands, on average, about 0.8σ away from its mean.

13.4 The empirical rule

Theorem 13.8. If $X \sim \text{Normal}(\mu, \sigma^2)$ then

$$\begin{aligned} \mathbb{P}\{\mu - \sigma \leq X \leq \mu + \sigma\} &\approx 68\%, \\ \mathbb{P}\{\mu - 2\sigma \leq X \leq \mu + 2\sigma\} &\approx 95\%, \\ \mathbb{P}\{\mu - 3\sigma \leq X \leq \mu + 3\sigma\} &\approx 99.7\%. \end{aligned}$$



Remark 13.9. This is worth internalizing because it turns σ into an interpretable number. If a process has $\mu = 60$ and $\sigma = 10$, then roughly 95% of output lies in $[40, 80]$, and a value of 100 is a four-sigma event — about 1 in 30,000. The rule is also the fastest sanity check on an exam: if your answer says $\mathbb{P}\{X > \mu + \sigma\} = 0.40$, you have made an arithmetic error, because it must be about 0.16.

Example 13.10 (How rare is a bad day?). A stock's daily return, in percent, is modeled as $\text{Normal}(0, 1)$, and a year has 252 trading days. How often should we see a one-day loss of more than 2%? Of more than 5%?

A loss of more than 2% is the event $Z < -2$, with probability $\Phi(-2) = 0.0228$; over a year that is $252 \times 0.0228 \approx 6$ days. A loss of more than 5% has probability $\Phi(-5) \approx 2.9 \times 10^{-7}$ (off the end of the usual table): about one day in 3.5 million, or once in roughly 14,000 years of trading.

Remark 13.11 (Fat tails). Stock markets have had many days with falls of more than 5% in the past forty years, and on October 19, 1987 the US market fell more than 20% in a single day. The normal model describes typical days well and extreme days badly: real returns have *fat tails*. The lesson is general. When a normal model says an event should happen once in ten thousand years and you have seen it several times, doubt the model, not your luck.

13.5 Looking ahead: the normal approximation to the binomial

If $X \sim \text{Binomial}(n, p)$ then $\mu = np$ and $\sigma^2 = np(1 - p)$. For large n the binomial pmf traces out a normal curve, and

$$\mathbb{P}\{X \leq x\} \approx \Phi\left(\frac{x + 0.5 - np}{\sqrt{np(1 - p)}}\right).$$

The $+0.5$ is the **continuity correction**: we are approximating a lattice-valued variable by a continuous one, so the discrete point x is represented by the interval $[x - 0.5, x + 0.5]$. The usual rule of thumb is that the approximation is good when $np \geq 10$ and $n(1 - p) \geq 10$.

We will see in Lecture 19 that this is just the central limit theorem applied to $X = \mathbf{1}_1 + \cdots + \mathbf{1}_n$, where $\mathbf{1}_i$ is the indicator of a success on the i -th trial: a binomial count is a sum of n i.i.d. terms.